

```
# Write a Program to input 2 numbers & print their sum.
```

```
a = int(input("Number: "))
b = int(input("Number: "))

print(a + b)
```

```
Number: 2
Number: 3
5
```

```
# WAP to input side of a square & print its area.
```

```
side_cm = int(input("side of square in cm: "))
print(side_cm * side_cm)
```

```
side of square in cm: 12
144
```

```
# WAP to input 2 floating point numbers & print their average.
```

```
a = float(input("Number: "))
b = float(input("Number: "))
print((a + b) / 2)
```

```
Number: 9
Number: 7.3
8.15
```

```
# WAP to input 2 int numbers, a and b.
```

```
# Print True if a is greater than or equal to b. If not print False.
```

```
a = int(input("Number: "))
b = int(input("Number: "))
def higher(a, b):
    output = False
    if a > b:
        output = True
    return output
print(higher(a,b))
```

```
Number: 6
Number: 3
True
```

```
# WAP to check if a number entered by the user is odd or even.
```

```
a = int(input("Number: "))
def odd_even(a):
    output = "odd"
    if int(a) % 2 == 0:
        output = "even"
    return output
print(odd_even(a))
```

```
Number: 3
odd
```

```
# WAP to find the greatest of 3 numbers entered by the user.
```

```
numbers =input("Enter your Numbers with a comma between each one:")
list_numbers = []
for x in numbers.split(","):
    list_numbers.append(int(x))
list_numbers.sort()

print("Largest", list_numbers[-1])
print("Second Largest",list_numbers[-2])
print("Third Largest", list_numbers[-3])
```

Enter your Numbers with a comma between each one:5,4,3,7,5,9
Largest 9
Second Largest 7
Third Largest 5

WAP to check if a number is a multiple of 7 or not.

```
x = int(input("Number: "))
if x % 7 == 0:
    print(True)
else:
    print(False)
```

Number: 8
False

WAP to ask the user to enter names of their 3 favorite movies & store them in a list.

```
words =input("Enter your 3 favourite movies with a comma between each one:")
list_words = []
for x in words.split(","):
    list_words.append(x)

print(list_words)
```

Enter your 3 favourite movies with a comma between each one:batman,spiderman,onepiece
['batman', 'spiderman', 'onepiece']

WAP to check if a list contains a palindrome of elements.
(Hint: use copy() method)

```
base_list = ["2", "apple", "apple", "2"]
reverse_list = base_list.copy()
reverse_list.reverse()
i = 0
palindrome = True

while i < len(base_list):
    if base_list[i] != reverse_list[i]:
        palindrome = False
        break
    i += 1

print(palindrome)
```

True

WAP to count the number of students with the "A" grade in the following tuple.

```
grades = ["C","D","A","A","B","B","A"]
i = 0
count = 0
while i < len(grades):
    if grades[i] == "A":
        count += 1
    i += 1
print(count)
```

Store the above values in a list & sort them from "A" to "D".

```
grades.sort()
print(grades)
```

3
['A', 'A', 'A', 'B', 'B', 'C', 'D']

```
# Store following word meanings in a python dictionary.
```

```
dictionary = {  
    "table": ["a piece of furniture", "list of facts & figures"],  
    "cat": "a small animal"  
}  
print(dictionary)
```

```
{'table': ['a piece of furniture', 'list of facts & figures'], 'cat': 'a small animal'}
```

```
# You are given a list of subjects for students.  
# Assume one classroom is required for 1 subject.  
# How many classrooms are needed by all students?  
# "python", "java", "C++", "python", "javascript",  
# "java", "python", "java", "C++", "C"
```

```
subjects = {  
    "python",  
    "java",  
    "C++",  
    "python",  
    "javascript",  
    "java",  
    "python",  
    "java",  
    "C++",  
    "C"  
}  
  
print(len(subjects))
```

```
5
```

```
# WAP to enter marks of 3 subjects from the user and store them in a dictionary.  
# Start with an empty dictionary & add one by one.  
# Use subject name as key & marks as value.
```

```
grades = {}  
  
for x in range(3):  
    subject = input("Enter your subject here please: ")  
    mark = int(input("Enter mark for subject: "))  
    grades[subject] = mark  
  
print(grades)
```

```
Enter your subject here please: maths  
Enter mark for subject: 98  
Enter your subject here please: english  
Enter mark for subject: 56  
Enter your subject here please: science  
Enter mark for subject: 87  
{'maths': 98, 'english': 56, 'science ': 87}
```

```
# Figure out a way to store 9 & 9.0 as separate values in the set.  
# (You can take help of built-in data types)
```

```
set = { 9, (9.0,)}  
print(set)
```

```
{9, (9.0,)}
```

```
# WAP to find the sum of first n numbers. (using while)
```

```
numbers = []  
n = int(input("How many numbers do you want to add: "))  
i = 0  
count = 0  
  
while i <= n:  
    count += i  
    i += 1  
  
print(count)
```

How many numbers do you want to add: 6
21

```
# WAP to find the factorial of first n numbers. (using for)

n = int(input("input n: "))
i = 1
output = 1
for x in range(n):
    output *= i
    i += 1

print(output)
```

input n: 3
6

Write a recursive function to calculate the sum of first n natural numbers.

```
def sum_n(n):
    if n == 0:
        return 0

    else:
        return n + sum_n(n-1)

n = int(input("Enter n: "))
print("Sum =", sum_n(n))
```

Enter n: 4
Sum = 10

Write a recursive function to print all elements in a list.
Hint : use list & index as parameters.

```
# Create a new file "practice.txt" using python.
# Add the following data in it:
# Hi everyone
# we are learning File I/O
# using Java.
# I like programming in Java.

f = open("practice.txt", "w")
f.write("Hi everyone\n")
f.write("we are learning File I/O\n")
f.write("using Java.\n")
f.write("I like programming in Java.\n")
f.close()

# WAF that replace all occurrences of "java" with "python".

with open("practice.txt", "r") as f:
    data = f.read()

# Replace Java with Python

data = data.replace("Java", "Python")

# Save changes

with open("practice.txt", "w") as f:
    f.write(data)

# Search if the word "learning" exists in the file or not.

if "learning" in data:
    print("Word found!")
```

```

    else:
        print("word not found")

# WAF to find in which line of the file does the word "learning" occur first.
# Print -1 if word not found.

def find_word_line(word):
    with open("practice.txt", "r") as f:
        line_num = 1

        for line in f:
            if word in line:
                return ("Found on line", line_num)
            line_num += 1
        return -1

print(find_word_line("learning"))

```

```

Word found!
('Found on line', 2)

```

```

# From a file containing numbers separated by comma,
# print the count of even numbers.

f = open("numbers.txt", "w")
f.write("6,7,2,77,98,3,4,65,66,42,78")
f.close()

with open("numbers.txt", "r") as f:
    data = f.read()

numbers = data.split(",")
count = 0

for num in numbers:
    if int(num) % 2 == 0:
        count += 1
print(count)

```

```

7

```

```

# Create student class that takes name & marks of 3 subjects
# as arguments in constructor.
# Then create a method to print the average.

class Student:
    def __init__(self, name, mark1, mark2, mark3):
        self.name = name
        self.mark1 = mark1
        self.mark2 = mark2
        self.mark3 = mark3

    def p_average(self):
        print("Average =", (self.mark1 + self.mark2 + self.mark3) / 3)

student1 = Student("Alice", 80, 90, 100)
student1.p_average()

```

```

Average = 90.0

```

```

# Create Account class with 2 attributes - balance & account no.

class Account:
    def __init__(self, balance, account_no):
        self.balance = balance
        self.account_no = account_no

account1 = Account( 2890, 54462218)
print(account1.balance)

def debit(self, n):
    self.balance = self.balance - n

def credit(self, n):
    self.balance = self.balance + n

def show_balance(self):

```

```
print(self.balance)

debit(account1, 100)
credit(account1, 200)
show_balance(account1)
```

```
2890
2990
```